

ABDUL RAZAK

Staff Data Scientist, Extreme Networks

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Portfolio | Google Scholar | DBLP | ORCID | Patents | LinkedIn

RESEARCH SUMMARY

Data scientist with more than a decade of experience taking machine-learning systems from prototype to production, backed by a research career in decision making under uncertainty. At Extreme Networks I lead the data-science work on generative models for network diagnosis. At Johnson Controls I built the probabilistic and rule-based models behind eight granted US patents in building security. The algorithms for multi-objective optimisation and preference inference that I developed during my doctorate and post-doctoral fellowship, published at AAI, IJCAI, UAI, CP and ICTAI with colleagues at IBM Research Dublin, still shape how I model trade-offs in production.

Research interests. Multi-objective decision making under uncertainty; preference elicitation and inference; probabilistic graphical models; machine learning; large language models and generative AI.

EDUCATION

2010–2014	PhD, Computer Science (Artificial Intelligence) National University of Ireland, Cork. Cork Constraint Computation Centre Thesis: <i>Reasoning with Imprecise Trade-offs in Decision Making under Certainty and Uncertainty</i> , supervised by Nic Wilson. Funded by the IRCSET / IBM Enterprise Partnership Scheme.
2003–2005	MSc, Mathematics with Computer Science Osmania University, Hyderabad, India Awarded the university merit scholarship.
2000–2003	BSc, Mathematics, Statistics and Computer Science Osmania University, Hyderabad, India

PROFESSIONAL APPOINTMENTS

Since 2020	Staff Data Scientist Extreme Networks, Scottsdale, Arizona ; previously Cork, Ireland - Lead the data-science work on generative models for network diagnosis, where language models generate and explain hypotheses over evidence drawn from anomaly detection, a knowledge graph and live telemetry. - Built agent workflows that triage client connectivity, roaming and device-health problems across thousands of access points and switches, and draft the remediation steps an operator applies in the field. - Developed correlation and inference pipelines over telemetry, device health and client-experience metrics, so that investigation of a routine fault now takes seconds rather than the hours it once demanded. - Adapted the preference-inference methods from my research to rank networking recommendations for customers whose priorities differ in reliability, cost and coverage across deployments of every size.
2017–2020	Data Scientist Johnson Controls, Cork, Ireland - Machine-learning lead for the intrusion-alarm, access-control and retail-analytics product lines. - Alarm-panel classification by Pareto optimisation, root-cause analysis of events, and site-independent signature generation for predictive security analysis. Eight of my granted US patents came from this work. - Relational probabilistic models in BLOG and Bayesian inference in PyMC3 across multi-site estates.
2014–2017	Post-Doctoral Fellow, Science Foundation Ireland Insight Centre for Data Analytics, NUI Cork - Probabilistic graphical models, clustering and influence diagrams for industrial decision problems, within a joint machine-learning research programme run with United Technology Research Centre and its partners. - Preference inference with lexicographic and weighted-average models, partly with IBM Research Dublin.
2005–2010	Lecturer in Mathematics Osmania University, Hyderabad, India Undergraduate teaching in algebra, real and numerical analysis, calculus, probability and statistics.

PEER-REVIEWED PUBLICATIONS

Listed most recent first. Complete record on [Google Scholar](#) and [DBLP](#).

- 2017 R. Marinescu, **A. Razak**, N. Wilson. Multi-objective influence diagrams with possibly optimal policies. *Proceedings of the 31st AAAI Conference on Artificial Intelligence*.
- 2015 Anne-Marie, **A. Razak**, N. Wilson. The comparison of multi-objective preference inference based on lexicographic and weighted average models. *Proceedings of the 27th IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*.
- 2015 N. Wilson, **A. Razak**, R. Marinescu. Computing possibly optimal solutions for multi-objective constraint optimisation with trade-offs. *Proceedings of the 24th International Joint Conference on Artificial Intelligence (IJCAI)*.
- 2015 Anne-Marie, **A. Razak**, N. Wilson. Multi-objective constraint optimisation with lexicographic preference models. *Workshop on Algorithmic Issues for Inference in Graphical Models (AIGM)*.
- 2013 R. Marinescu, **A. Razak**, N. Wilson. Multi-objective constraint optimisation with trade-offs. *Proceedings of the 19th International Conference on Principles and Practice of Constraint Programming (CP)*.
- 2012 R. Marinescu, **A. Razak**, N. Wilson. Multi-objective influence diagrams with trade-offs. *Proceedings of the 28th Conference on Uncertainty in Artificial Intelligence (UAI)*.

GRANTED US PATENTS

Nine granted between 2019 and 2021, grouped by subject. Filings are searchable on [Justia](#) and [Google Patents](#).

Decision intelligence and maintenance

- 2019 **A. Razak**, L. A. Deleris, R. Marinescu, N. Wilson. Interactive method to reduce the amount of trade-off information required from decision-maker in multi-attribute decision-making under uncertainty.
- 2019 **A. Razak**, G. Subramanian, C. J. Donovan. Virtual maintenance manager.
- 2021 D. Horgan, **A. Razak**, G. Bernardi. Root-cause analysis of event occurrences.

Alarm management

- 2020 **A. Razak**, M. Stewart, G. Subramanian. Automated alarm panel classification using Pareto optimization.
- 2021 **A. Razak**. Systems and methods for managing alarm data of multiple locations.
- 2021 M. Costello, F. Fala, J. R. Holliday, N. B. Pavel, **A. Razak**. Building security system with alarm lists and alarm rule editing.

Building security analytics

- 2020 M. Dziduch, C. J. Donovan, **A. Razak**. Building security analysis system with site-independent signature generation for predictive security analysis.
- 2020 M. Dziduch, C. J. Donovan, **A. Razak**. Rules-based method of identifying misuse of emergency fire exits using data generated by a security alarm system.
- 2020 M. Dziduch, C. J. Donovan, **A. Razak**. Systems and methods for managing a security system.

HONOURS AND FUNDING

- IEEE Senior Member, elevated May 2026.
- Science Foundation Ireland post-doctoral fellowship, Insight Centre for Data Analytics, NUI Cork, 2014 to 2017.
- IRCSET / IBM Enterprise Partnership Scheme scholarship, full doctoral funding, 2010 to 2013.
- Osmania University merit scholarship (MSc, 2003 to 2005), and 40th place of roughly 30,000 candidates in the university mathematics entrance examination.

ACADEMIC SERVICE AND INVITED TALKS

- Reviewer, *Computers and Electrical Engineering* (Elsevier). More than ten manuscripts in artificial intelligence, wireless networking and intelligent systems.
- Reviewer, IEEE International Conference on Communications (ICC), artificial-intelligence submissions to the 2025 technical programme.
- Invited talks: Sapienza University of Rome, on preference handling in decision making under certainty and uncertainty; NUI Galway, on probabilistic decision making under uncertainty; NUI Cork and IBM Research Dublin, during the Risk Management Collaboratory and United Technology Research Centre programmes.
- Conference presentations, oral and poster, at AAAI, IJCAI, UAI, CP and ICTAI.
- Five years of undergraduate lecturing, and mentoring of junior data scientists in industry.

TECHNICAL SKILLS

Programming	Python, C++ (STL, Boost, Eigen), SQL, PySpark, PyFlink.
Modelling	PyMC3, BLOG, Swift, ILOG CPLEX, LPSOLVE, scikit-learn.
Platforms	AWS SageMaker and Athena, Azure Databricks, streaming and batch pipelines.
Languages	English, Hindi, Telugu, Urdu. Irish citizen since 2021.